# Amazon Sales Performance Analysis & Dashboard

Data Analytics Project using Python, Pandas & Power BI

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# Project Overview :

This project presents a comprehensive analysis of Amazon store sales data to understand business performance, customer behavior, and profitability trends. The dataset consists of key attributes such as sales, profit, product categories, customer segments, shipping modes, and regional information, which are essential for evaluating overall business operations. Using data analytics techniques and visualization tools like Power BI, the project explores patterns in sales growth, identifies seasonal fluctuations, and analyzes category-wise performance across different time periods. Additionally, the study highlights variations in profitability among products and regions, helping to identify areas of improvement. The main objective of this analysis is to derive actionable insights that support data-driven decision-making, enhance operational efficiency, and improve overall business performance.

# Project Objective :

- Analyze overall sales performance and identify key trends in revenue and profit
- Understand customer behavior across different segments, regions, and purchasing patterns
- Evaluate product category performance to identify high-performing and low-performing areas
- Examine the impact of shipping modes and payment methods on sales and profitability
- Identify seasonal trends and variations in sales over time
- Develop an interactive dashboard using Power BI for clear and effective data visualization

# Problem Statement :

- Sales and profit are not consistently aligned, with some transactions generating high revenue but low or negative profit
- The presence of negative profit values indicates loss-making products and potential inefficiencies in pricing or cost management
- Sales distribution is highly skewed, where a small number of transactions contribute significantly to total revenue
- Customer demand and order quantities vary across different segments and regions, making it difficult to identify stable patterns
- Profitability differs across product categories, sub-categories, and regions, indicating uneven business performance
- Identifying top-performing and loss-making customers is essential for improving overall business strategy

# Summary Statistics :

The dataset shows that average sales are around 265, while the median is lower, indicating a right-skewed distribution with some high-value transactions. Profit also exhibits high variability, with a significant negative minimum value (-6599), highlighting the presence of loss-making transactions. The large difference between minimum and maximum values in both sales and profit suggests the existence of outliers and uneven distribution. Overall, the data demonstrates high variability across transactions.

```
df.describe()
```

| | Sales | Quantity | Profit | Returns |
|---|---|---|---|---|
| count | 5901.000000 | 5901.000000 | 5901.000000 | 5901.000000 |
| mean | 265.345589 | 3.781901 | 29.700408 | 0.048636 |
| std | 474.260645 | 2.212917 | 259.589138 | 0.215124 |
| min | 0.836000 | 1.000000 | -6599.978000 | 0.000000 |
| 25% | 71.976000 | 2.000000 | 1.795500 | 0.000000 |
| 50% | 128.648000 | 3.000000 | 8.502500 | 0.000000 |
| 75% | 265.170000 | 5.000000 | 28.615000 | 0.000000 |
| max | 9099.930000 | 14.000000 | 8399.976000 | 1.000000 |

# Dataset Overview :

- The dataset contains 5901 records and 20 columns representing Amazon sales transactions
- It includes key attributes such as Order ID, Customer Name, Segment, Region, Category, and Sub-Category
- Important numerical features include Sales, Quantity, Profit, and Returns
- The dataset covers multiple regions and customer segments for comprehensive analysis

```
•••  <class 'pandas.core.frame.DataFrame'>
     Index: 5901 entries, 4918 to 5093
     Data columns (total 20 columns):
      #   Column         Non-Null Count  Dtype
     ---  ------         --------------  -----
      0   Order ID       5901 non-null   object
      1   Order Date     5901 non-null   object
      2   Ship Date      5901 non-null   object
      3   Ship Mode      5901 non-null   object
      4   Customer ID    5901 non-null   object
      5   Customer Name  5901 non-null   object
      6   Segment        5901 non-null   object
      7   Country        5901 non-null   object
      8   City           5901 non-null   object
      9   State          5901 non-null   object
      10  Region         5901 non-null   object
      11  Product ID     5901 non-null   object
      12  Category       5901 non-null   object
      13  Sub-Category   5901 non-null   object
      14  Product Name   5901 non-null   object
      15  Sales          5901 non-null   float64
      16  Quantity       5901 non-null   int64
      17  Profit         5901 non-null   float64
      18  Returns        5901 non-null   int64
      19  Payment Mode   5901 non-null   object
     dtypes: float64(2), int64(2), object(16)
     memory usage: 968.1+ KB
```

| | Order ID | Order Date | Ship Date | Ship Mode | Customer ID | Customer Name | Segment | Country | City | State | Region | Product ID | Category | Sub-Category | Product Name | Sales | Quantity | Profit | Returns | Payment Mode |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Row O6G3A1:R6 | | | | | | | | | | | | | | | | | | | | |
| 4918 | CA-2019-160304 | 1/1/2019 | 1/7/2019 | Standard Class | BM-11575 | Brendan Murry | Corporate | United States | Gaithersburg | Maryland | East | FUR-BO-10004709 | Furniture | Bookcases | Bush Westfield Collection Bookcases, Medium Ch... | 73.94 | 1 | 28.2668 | 0 | Online |
| 4919 | CA-2019-160304 | 1/2/2019 | 1/7/2019 | Standard Class | BM-11575 | Brendan Murry | Corporate | United States | Gaithersburg | Maryland | East | FUR-BO-10004709 | Furniture | Bookcases | Bush Westfield Collection Bookcases, Medium Ch... | 173.94 | 3 | 38.2668 | 0 | Online |
| 4920 | CA-2019-160304 | 1/2/2019 | 1/7/2019 | Standard Class | BM-11575 | Brendan Murry | Corporate | United States | Gaithersburg | Maryland | East | TEC-PH-10000455 | Technology | Phones | GE 30522EE2 | 231.98 | 2 | 67.2742 | 0 | Cards |
| 3074 | CA-2019-125206 | 1/3/2019 | 1/5/2019 | First Class | LR-16915 | Lena Radford | Consumer | United States | Los Angeles | California | West | OFF-ST-10003692 | Office Supplies | Storage | Recycled Steel Personal File for Hanging File ... | 114.46 | 2 | 28.6150 | 0 | Online |
| 8604 | US-2019-116365 | 1/3/2019 | 1/8/2019 | Standard Class | CA-12310 | Christine Abelman | Corporate | United States | San Antonio | Texas | Central | TEC-AC-10002217 | Technology | Accessories | Imation Clip USB flash drive - 8 GB | 30.08 | 2 | -5.2640 | 0 | Online |

| | Order ID | Order Date | Ship Date | Ship Mode | Customer ID | Customer Name | Segment | Country | City | State | Region | Product ID | Category | Sub-Category | Product Name | Sales | Quantity | Profit | Returns | Payment Mode |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Row G3A1:R6 | | | | | | | | | | | | | | | | | | | | |
| 907 | CA-2020-143259 | 12/30/2020 | 1/3/2021 | Standard Class | PO-18865 | Patrick O'Donnell | Consumer | United States | New York City | New York | East | FUR-BO-10003441 | Furniture | Bookcases | Bush Westfield Collection Bookcases, Fully Ass... | 213.136 | 4 | 12.1176 | 0 | COD |
| 1297 | CA-2020-115427 | 12/30/2020 | 1/3/2021 | Standard Class | EB-13975 | Erica Bern | Corporate | United States | Fairfield | California | West | OFF-BI-10002103 | Office Supplies | Binders | Cardinal Slant-D Ring Binder, Heavy Gauge Vinyl | 295.904 | 2 | 4.5188 | 1 | Online |
| 5092 | CA-2020-156720 | 12/30/2020 | 1/3/2021 | Standard Class | JM-15580 | Jill Matthias | Consumer | United States | Loveland | Colorado | West | OFF-FA-10003472 | Office Supplies | Fasteners | Bagged Rubber Bands | 388.024 | 3 | -0.6048 | 0 | Online |
| 909 | CA-2020-143259 | 12/30/2020 | 1/3/2021 | Standard Class | PO-18865 | Patrick O'Donnell | Consumer | United States | New York City | New York | East | OFF-BI-10003684 | Office Supplies | Binders | Wilson Jones Legal Size Ring Binders | 462.776 | 3 | 19.7910 | 0 | COD |
| 5093 | CA-2020-151450 | 12/31/2020 | 1/4/2021 | Standard Class | JM-15580 | Jill Matthias | Consumer | United States | Loveland | Colorado | West | OFF-FA-10003472 | Office Supplies | Fasteners | Bagged Rubber Bands | 2975.510 | 3 | -0.6048 | 0 | COD |

# Data Cleaning :

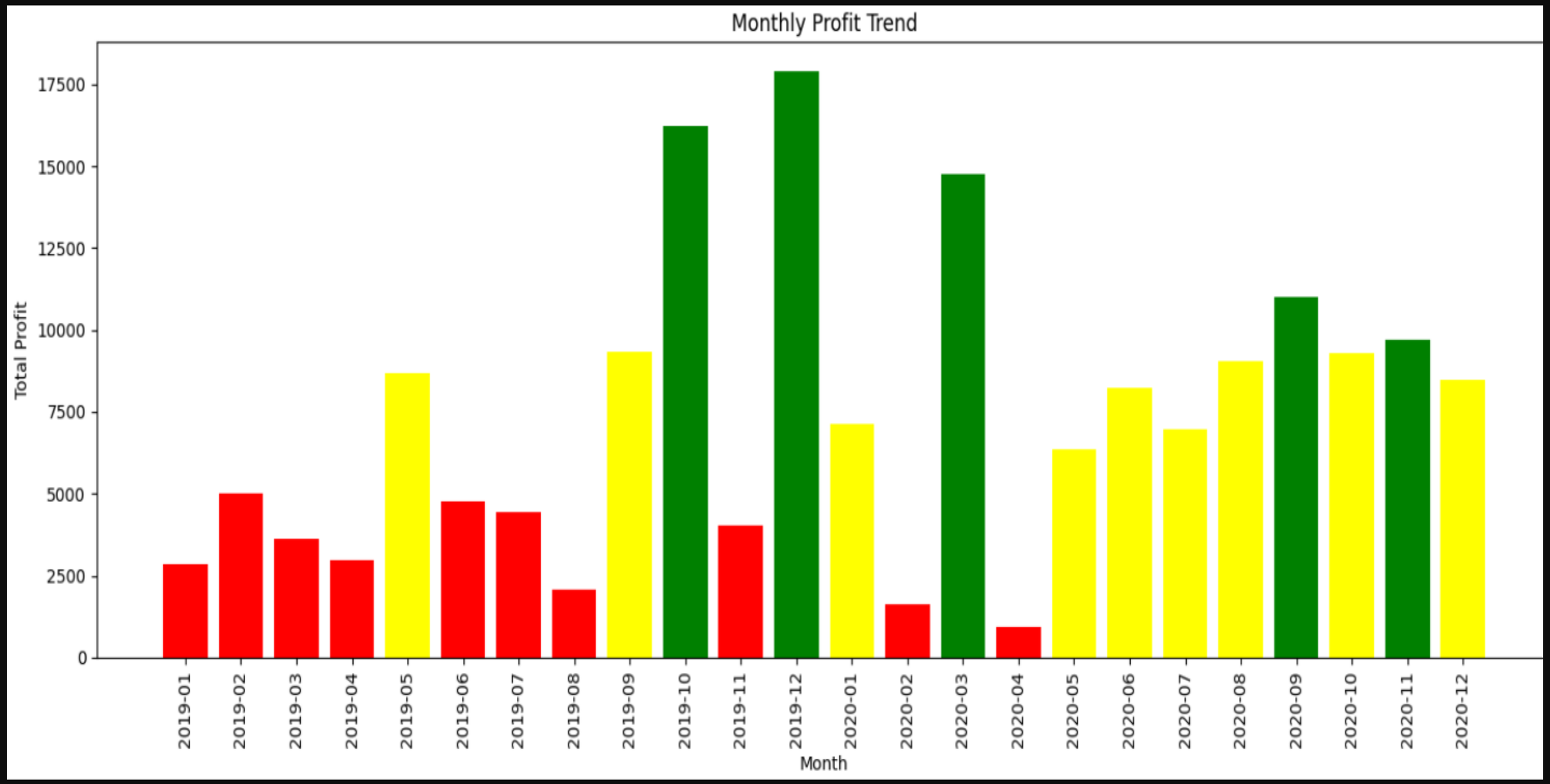
- The data quality and reliability, data cleaning and validation steps were performed on the dataset.
- Verified that the dataset contains no missing values or duplicate records
- Ensured correct data types for numerical and categorical features
- Identified customers with negative profit values indicating loss-making transactions
- Profit < 0 highlights cases where cost exceeds revenue (Profit = Sales – Cost)
- These records were analyzed to understand patterns affecting overall profitability

| | Customer ID | Customer Name | Profit |
|---|---|---|---|
| 177 | CS-12505 | Cindy Stewart | -6652.7235 |
| 303 | GT-14635 | Grant Thornton | -3825.3394 |
| 692 | SR-20425 | Sharelle Roach | -3391.4082 |
| 314 | HG-14965 | Henry Goldwyn | -2835.2529 |
| 541 | NC-18415 | Nathan Cano | -2251.0832 |
| ... | ... | ... | ... |
| 741 | TS-21085 | Thais Sissman | -1.9656 |
| 335 | JC-15340 | Jasper Cacioppo | -1.6813 |
| 495 | MG-18205 | Mitch Gastineau | -1.2453 |
| 245 | EB-13930 | Eric Barreto | -0.7438 |
| 598 | PW-19240 | Pierre Wener | -0.5904 |

147 rows × 3 columns

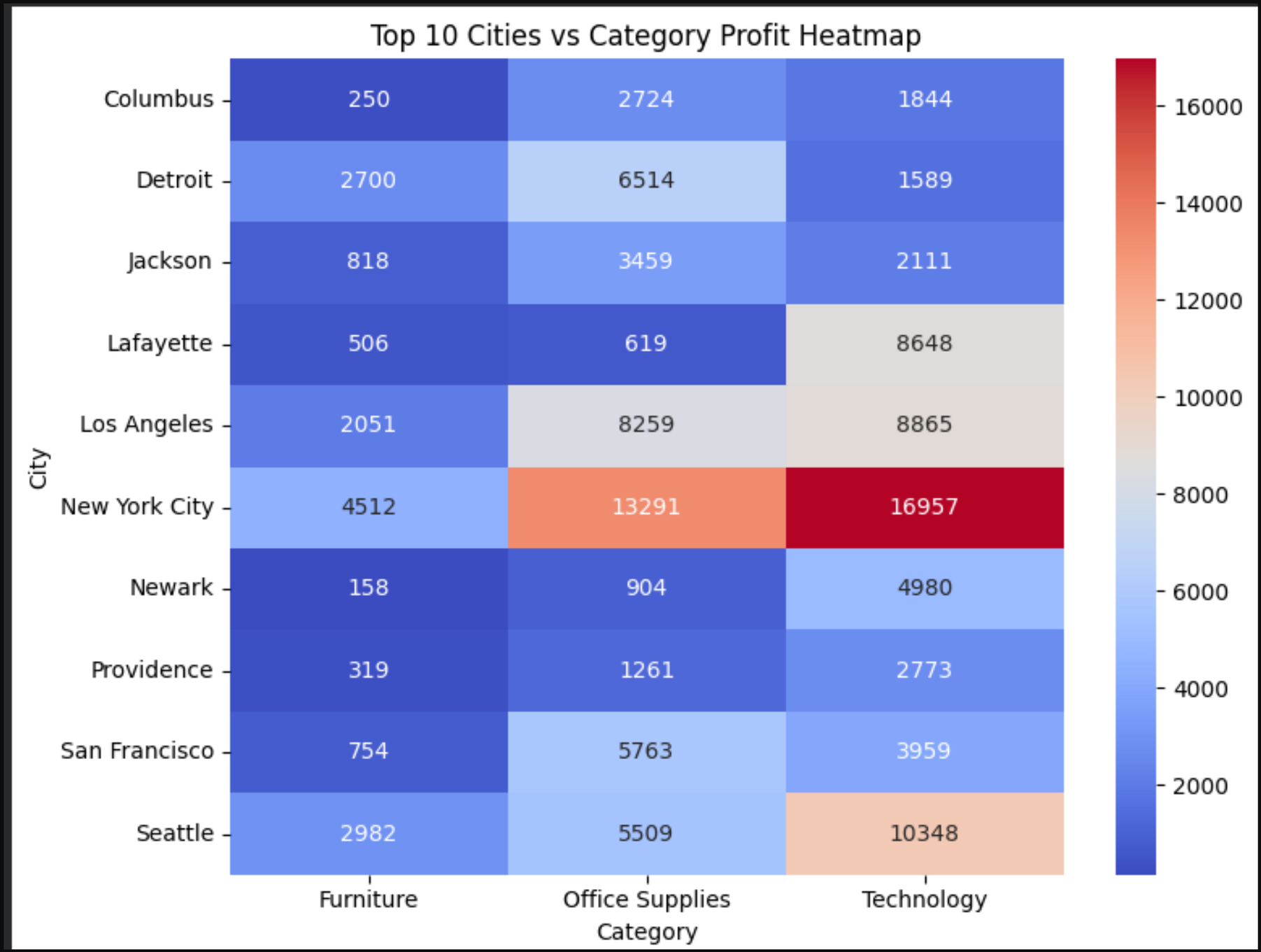
# Monthly Profit Analysis :

- **Color Indicator:** Red represents low profit, yellow indicates medium performance, and green shows high profit levels
- **Trend Overview:** Profit fluctuates across months, showing inconsistent performance over time
- **Peak Performance:** Green bars in late 2019 highlight the highest profit periods
- **Low Performance:** Red bars in early 2020 indicate a sharp decline in profit
- **Recovery Phase:** A gradual shift from red/yellow to green in later months shows recovery and improvement
- **Overall Insight:** Color patterns clearly reflect seasonal trends and changing business performance


Monthly Profit Trend
Total Profit
17500
15000
12500
10000
7500
5000
2500
0
2019-01
2019-02
2019-03
2019-04
2019-05
2019-06
2019-07
2019-08
2019-09
2019-10
2019-11
2019-12
2020-01
2020-02
2020-03
2020-04
2020-05
2020-06
2020-07
2020-08
2020-09
2020-10
2020-11
2020-12
Month

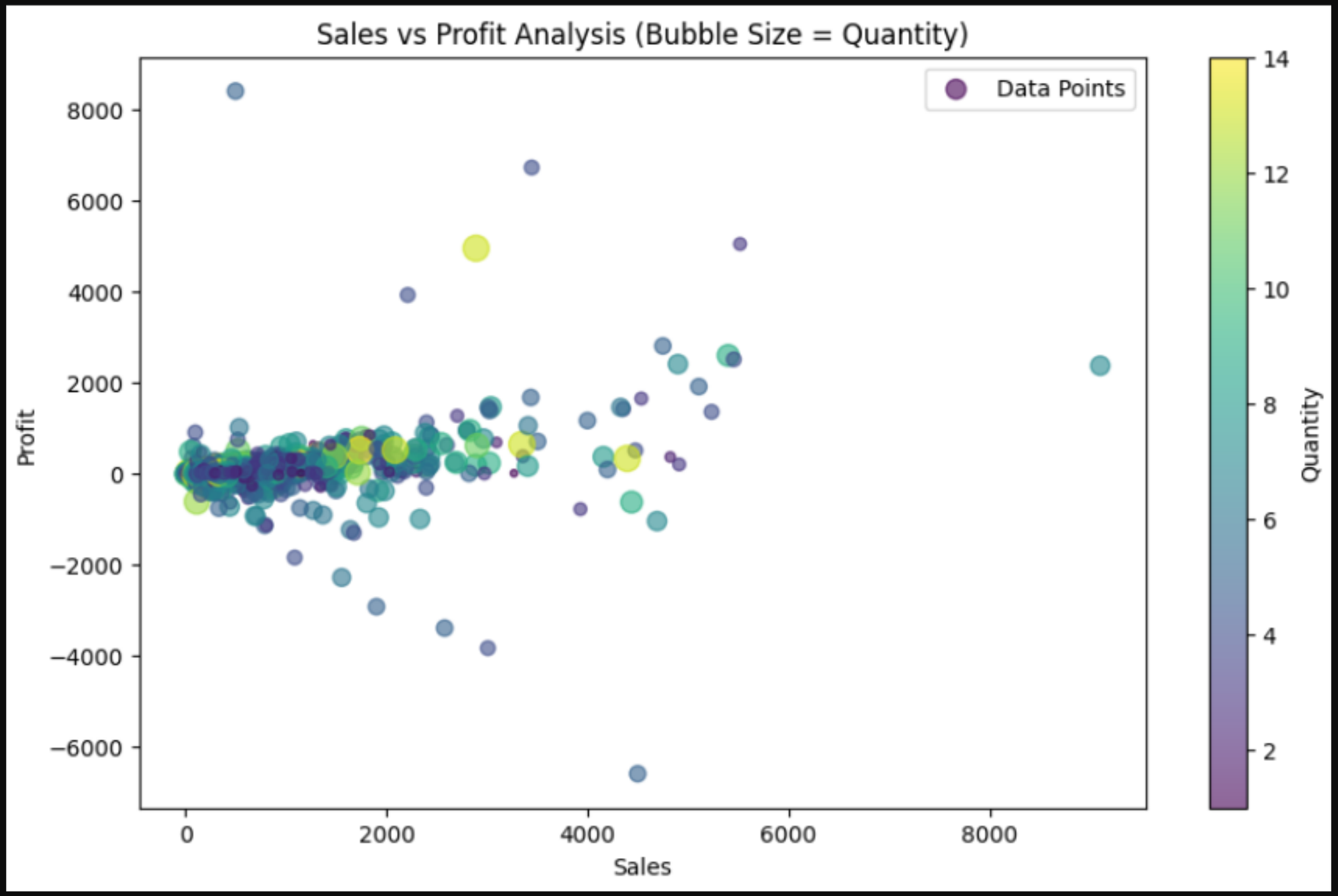
# Profit Analysis by City and Category :

- New York City generates the highest profit, with Technology (16957) and Office Supplies (13291), making it the top-performing city overall
- Technology is the most profitable category, with high values in cities like Seattle (10348), Los Angeles (8865), and Lafayette (8648)
- Office Supplies shows consistent performance across cities, with profits such as Los Angeles (8259) and San Francisco (5763)
- Furniture contributes comparatively lower profit, with several cities below 3000, such as Columbus (250) and Providence (319)
- Cities like Columbus and Newark show low overall performance, with profits below 3000 across most categories
- The large gap between highest profit (16957) and lowest profit (158) highlights significant variation in profitability across cities


Top 10 Cities vs Category Profit Heatmap
Columbus
Detroit
Jackson
Lafayette
Los Angeles
New York City
Newark
Providence
San Francisco
Seattle
City
250
2724
1844
2700
6514
1589
818
3459
2111
506
619
8648
2051
8259
8865
4512
13291
16957
158
904
4980
319
1261
2773
754
5763
3959
2982
5509
10348
Furniture
Office Supplies
Technology
Category
16000
14000
12000
10000
8000
6000
4000
2000

# Profit Drivers Analysis :

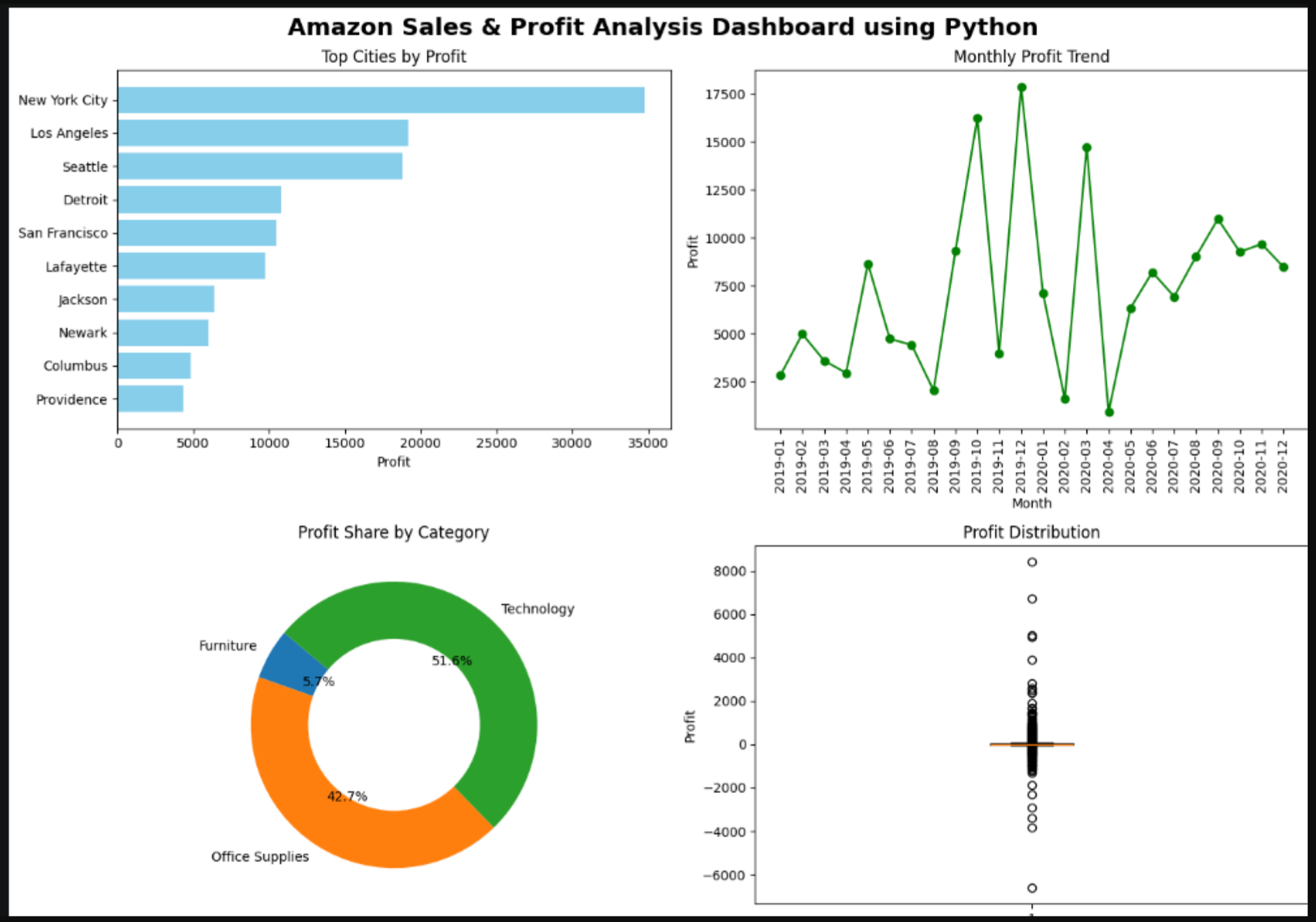
- **Relationship Trend:** Sales and profit show a weak to moderate positive relationship, as higher sales generally lead to higher profit but not consistently
- **Profit Variability:** Profit ranges widely from around -6500 to 8000, indicating both high losses and high gains across transactions
- **Loss Region:** Several data points with moderate sales (2000–4000) show negative profit, highlighting inefficiencies or high costs
- **Quantity Impact:** Larger bubbles (higher quantity) are mostly concentrated in mid-sales ranges, but do not always correspond to higher profit
- **Color Representation:** Dark colors (low quantity) and bright yellow-green colors (high quantity) indicate that higher quantity does not guarantee higher profitability
- **Outliers:** A few high-profit points (>7000) exist at moderate sales levels, indicating highly profitable transactions
- **Overall Insight:** Profitability depends not only on sales and quantity but also on pricing, cost, and product mix


Sales vs Profit Analysis (Bubble Size = Quantity)
Data Points
Profit
Sales
Quantity
8000
6000
4000
2000
0
−2000
−4000
−6000
0
2000
4000
6000
8000
14
12
10
8
6
4
2

# Sales & Profit Insights :

- **Top Performing Cities:** New York City generates the highest profit, followed by Los Angeles and Seattle, making them key revenue contributors
- **Monthly Profit Trend:** Profit shows fluctuations over time, with noticeable peaks in late 2019 and mid-2020, indicating seasonal variation
- **Category Contribution:** Technology leads with 51.6% profit share, followed by Office Supplies (42.7%), while Furniture contributes only 5.7%
- **Profit Distribution:** The presence of both high positive values and negative outliers indicates high variability in profit across transactions
- **Business Insight:** Profitability varies across cities, categories, and time, highlighting multiple factors influencing business performance

# Dashboard :


Amazon Sales & Profit Analysis Dashboard using Python
Top Cities by Profit
New York City
Los Angeles
Seattle
Detroit
San Francisco
Lafayette
Jackson
Newark
Columbus
Providence
0
5000
10000
15000
20000
25000
30000
35000
Profit
Monthly Profit Trend
17500
15000
12500
10000
7500
5000
2500
Profit
2019-01
2019-02
2019-03
2019-04
2019-05
2019-06
2019-07
2019-08
2019-09
2019-10
2019-11
2019-12
2020-01
2020-02
2020-03
2020-04
2020-05
2020-06
2020-07
2020-08
2020-09
2020-10
2020-11
2020-12
Month
Profit Share by Category
Technology
Furniture
51.6%
5.7%
42.7%
Office Supplies
Profit Distribution
8000
6000
4000
2000
0
−2000
−4000
−6000
Profit

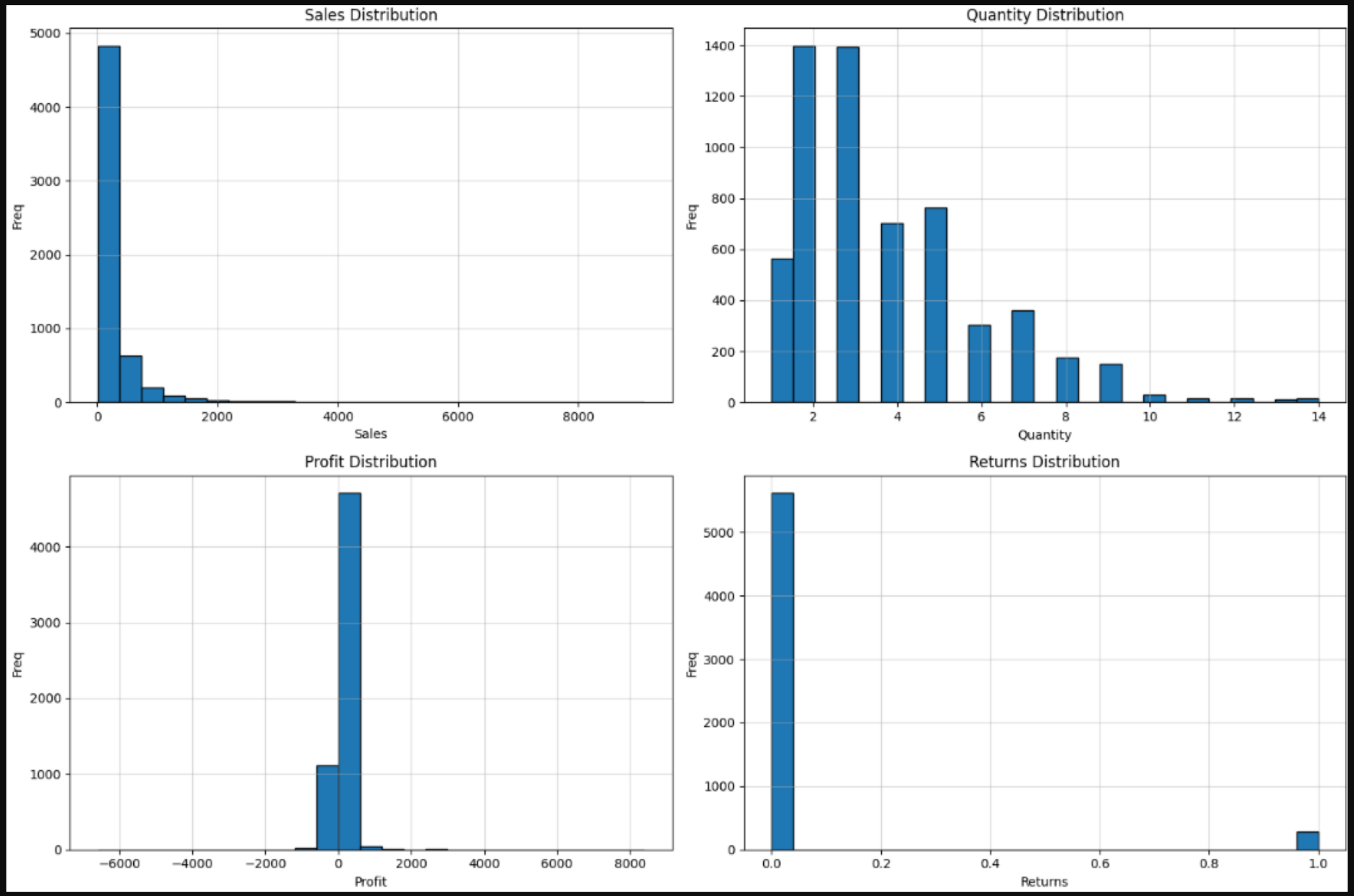
# Data Distribution Analysis:

I. **Sales Distribution:** Sales values are highly concentrated at the lower end, with most transactions occurring below 500. Only a few values extend beyond 2000, indicating the presence of high-value outliers. This creates a right-skewed distribution where small sales dominate the dataset

I. **Quantity Distribution:** The majority of orders involve small quantities ranging from 1 to 5 units. Higher quantities above 10 are rarely observed, indicating that bulk purchases are uncommon. This reflects a pattern of frequent low-volume transactions

II. **Profit Distribution:** Profit values vary widely, ranging from approximately -6500 to 8000. However, most transactions are clustered near 0 to 500, showing moderate profitability. The presence of negative values indicates that some transactions result in losses

III. **Returns Distribution:** Most values are concentrated at 0, indicating that returns are very rare in the dataset. Only a few instances reach a value of 1, suggesting minimal return activity. This highlights overall stability in order fulfillment


Sales Distribution
Freq
Sales
Quantity Distribution
Freq
Quantity
Profit Distribution
Freq
Profit
Returns Distribution
Freq
Returns

# Power BI Dashboard Overview:

This dashboard represents a complete end-to-end analysis of the Amazon sales dataset, where I performed data cleaning, transformation, and exploratory analysis using Python (Pandas, NumPy) before developing an interactive dashboard in Power BI.

The solution integrates key business KPIs such as total sales, profit, order volume, and product performance to provide a comprehensive view of operations. I designed multiple visualizations to analyze trends over time, category-wise performance, regional distribution, and customer behavior.

DAX functions were implemented to create calculated measures and enhance analytical depth, enabling dynamic insights. Interactive filters and slicers allow users to drill down into specific regions, states, and segments for detailed analysis.

The dashboard helps identify high-performing products, detect loss-making areas, and understand seasonal trends and variability in performance.

It supports efficient monitoring of business metrics and improves decision-making by providing clear, data-driven insights. Overall, this project demonstrates my ability to transform raw data into structured, scalable, and actionable business intelligence solutions.

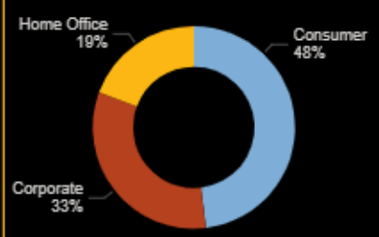
# AMAZON STORE SALES DASHBOARD

Central | East | South | West

Choose State

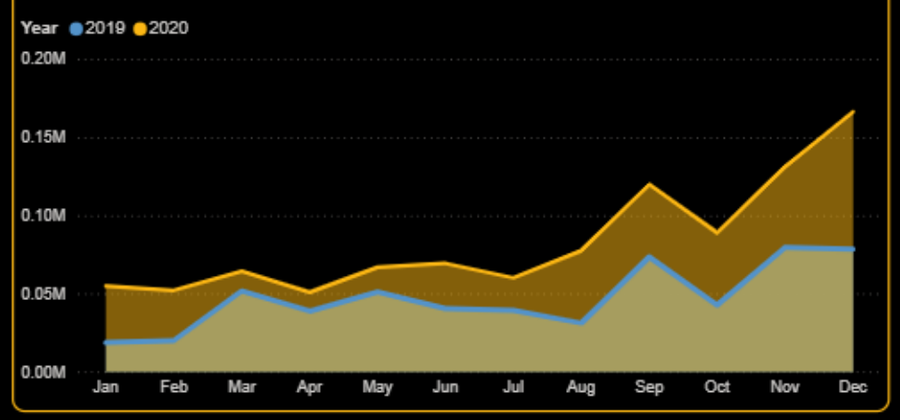
All

## Sales by Segment

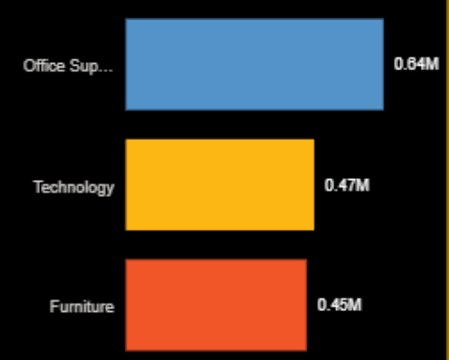



| Sales | Profit | Orders | Total Products |
|---|---|---|---|
| 1.57M | 175K | 5901 | 1755 |

## Seasonal Fluctuation of Sales Per Year

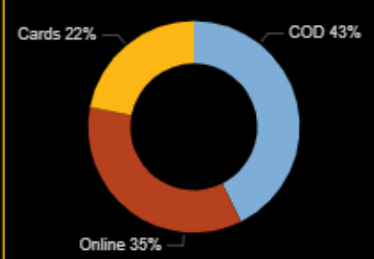



## Sales by Category

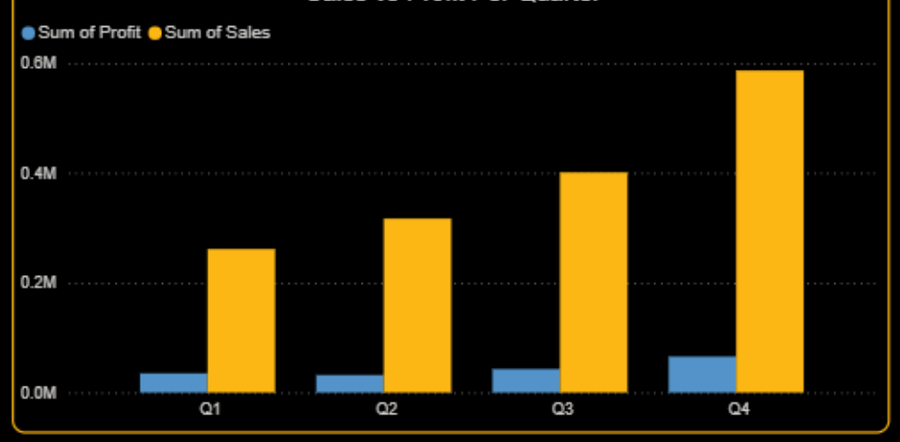



## Sum of Sales by State

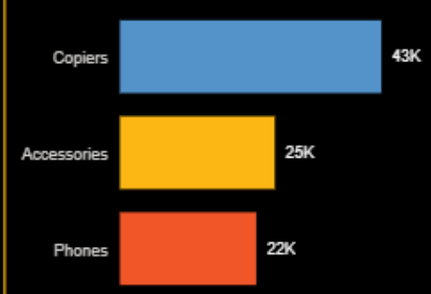
## Sales by Payment Mode



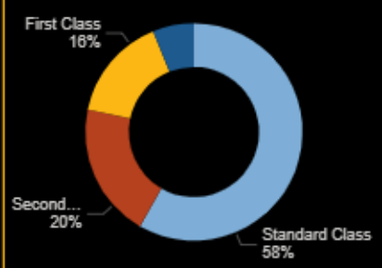

## Sales vs Profit Per Quarter



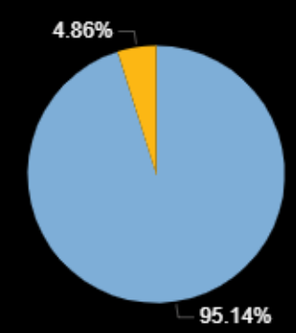

## Top 3 Profitable Sub-Categories




## Sales by Ship Mode




## Return Status



# Project Outcomes

This analysis focuses on understanding sales performance, profitability patterns, and customer behavior within the Amazon dataset. The exploration highlights how profit varies across cities, product categories, and time, showing that higher sales volume does not always guarantee higher profitability. Key patterns such as strong performance in the Technology category, regional differences in profit contribution, and fluctuations in monthly trends were identified. The analysis also reveals the presence of loss-making transactions and variability in profit distribution, emphasizing the impact of pricing, cost, and product mix. The developed dashboard transforms these insights into an interactive format, enabling quick identification of trends, performance gaps, and opportunities. This approach enhances decision-making by providing a clear and data-driven understanding of business performance.

**Thank You for Your Time 😊**